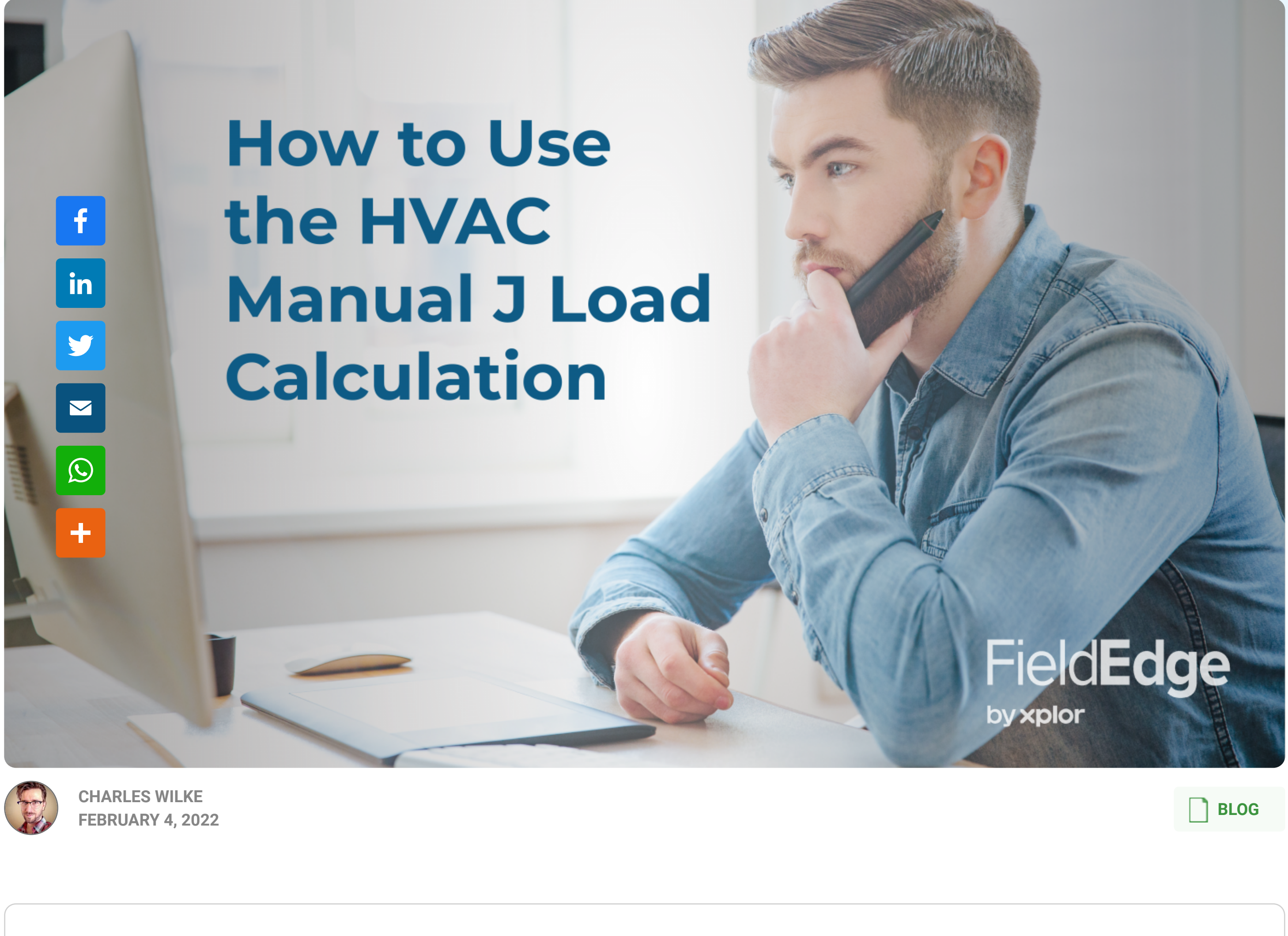




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BLOG

How to Use the HVAC Manual J Load Calculation

Understanding how to heat and cool a building is a central skill in the HVAC industry. The Manual J Load calculation helps determine the size required for an HVAC unit to sustain the environmental temperature of a home.

This article will get you up to speed on the following:

- An overview of the Manual J Load calculation.
- An in-depth look at the variables that go into this calculation.
- The benefits this offers your HVAC customers and how it impacts their satisfaction.

What Is the Manual J Load Calculation?

Also known as the Residential Load Calculation, the **Manual J Load calculation** was developed by the **Air Conditioning Contractors of America** (ACCA).

“[This calculation] is a protocol contractors use to determine a home’s precise heating and cooling requirements,” **notes** Modesto, California-based Derek Sawyers Heating & Air.

This calculation offers a more tailored approach to defining the overall heating and cooling needs of a building, compared to the classic **rule-of-thumb** approach.

These rules of thumb often led to buildings receiving recommendations for larger than necessary HVAC systems. The Manual J Load calculation was developed to benefit customers with a more individualized solution per building, saving money and **satisfying customers**.

Variables Within the Manual J Load Calculation

ACCA provides a handy overview of the variables that go into defining the values within this calculation:



While the basics of the calculation are straightforward, there is some nuance to understanding the finer variables.

Aside from the square footage and ceiling height, an understanding on the current state of insulation, amount of windows and entryways, as well as the type of windows all play into the viability of this calculation.

Other important variables include:

Occupant Count

Perhaps the most important variable is knowing how many people occupy the space. Each person would add 100 BTUs to the total necessary HVAC system output, which might seem like a lot.

However, the standard rule of thumb is to add 1,000 BTUs for every exterior door! For reference, 12,000 BTUs equals one ton. This is a useful conversion to keep in mind when judging the output of any HVAC system.

Exterior Doors

It’s not as simple as counting doors to understand how these will affect the overall Manual J Load calculation. Since newer doors are usually more tightly sealed than those decades old. Check for drafts with closed exterior doors.

While there are some easy-to-purchase solutions to a drafty door, like weather stripping, this variable will impact the overall efficacy of an entire HVAC system on a building.

Walls and Partitions

Any division of spaces will affect how an HVAC system conditions the air. Include **pony walls**, or half walls, as well.

Floors and Foundation

The floor’s materials affect the total efficacy of an HVAC system as well as its relationship with the foundation of the structure. If the home is built on a slab, then that will affect this calculation differently than if the floor is elevated from the foundation.

An air cushion between the interior floor and the ground will create a natural insulator.

Also, if the floor is below grade, meaning beneath the surface of the ground outside, this also plays a factor. Depending on the depth below grade, the floor may actually benefit from some natural insulation.

All Things Windows

It’s obvious windows have a direct effect on the temperature of a space. Although, not all windows are created equal. There’s four main characteristics that play into this calculation:

1. Glazing
2. Direction
3. Fenestration
4. Number of windows

For example, let’s say there’s three big windows that let in the afternoon sun. The glazing, or type and amount of panes that make up each window will affect their overall insulation quotient. Triple pane windows create that air buffer between panes which make excellent insulators.

Also, the fenestration of these windows refers to the overall shape. Are these tall slit windows that let in a narrow band of light? Or are these picture frame windows that let the light pour in?

Climate Region

Keep in mind, there’s six major **climate regions** within the U.S.

Depending on where the home is located, you’ll need to adjust your calculations.

Insulation Type and Use of Space

Insulation is measured by its **R-Value**. This value is necessary when completing your calculations.

While insulation is important, the use of the space factors into the overall consideration for HVAC system size and output. A busy kitchen with people flowing in and out every day is going to stay warmer than a storage room. Factor this in as you calculate.

You can see in **this example** from ACCA how all these factors work together to create a presentable recommendation.

Customer Benefits from the Manual J Load Calculation

Since this calculation gives a much more tailored view of the needs of each HVAC customer, they benefit greatly from its use.

Customer benefits include:

- Tailored solution that fits their individual needs and use case
- Saving money, not just on equipment and install, but also in daily use
- Peace of mind knowing the HVAC solution will be well equipped to handle temperature control of the space
- The personal satisfaction in knowing the HVAC company they choose has the most current mathematical tools to base their recommendation
- Less maintenance and repairs over time since the HVAC system chosen is built to accommodate the space and its use

Needless to say, showing the math on your sales recommendations will go a long way in proving your commitment to personalized care.

Ramp Up Your Manual J Load Calculation Efforts With the #1 HVAC Software

With this calculation in hand, your field team will have the necessary data to suggest the best possible HVAC solution that’s tailored to each customer’s unique needs.

- Include the Manual J Load value in your **customer profiles** for easy access in the office and the field, in case an equipment replacement is needed in the future.
- Promote the use of this calculation in your **email marketing** to drive customer awareness.
- vSince your customers will feel the tailored touch of your recommendations, annual **service agreements** become a snap to sell.

See how FieldEdge can help you maximize your Manual J Load Calculation with good, better, best field price presentations. Book your FREE personalized demo today!

Book Demo

You Can’t Afford NOT to Use the Manual J Load Calculation

As an HVAC company, you have a commitment to satisfying your customers. Perhaps the most integral way to do that is by providing them the professional recommendations that are right for them and their space.

Using the Manual J Load calculation, along with a personal walkthrough and consultation, you’ll ensure your customers are only spending on the hardware they need to properly control the climate of their home.

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